Quiz 2

1.

After interviews with patients, doctors, receptionists, and hospital administrators, the following six requirements were identified:

- **Patients should be able to search for doctors by specialty and book available appointment slots.**
- **The system should allow patients to pay consultation fees using debit cards, credit cards, mobile banking, or digital wallets.**
- **Appointment confirmations should be sent to patients within 5 seconds after successful booking.**
- **The system should support at least 20,000 concurrent users during peak hours without noticeable performance degradation.**
- **Doctors should receive real-time notifications when patients upload medical reports before their appointments.**
- **The system should maintain at least 99.9% uptime and scheduled maintenance downtime should not exceed 2 hours per month.**

a) For each of the above six requirements, classify them as either Functional or Non-Functional, and justify your answer with one sentence each.

1. **Functional Requirement.** It describes a feature that allows patients to search for doctors by specialty and book available appointment slots.
2. **Functional Requirement.** It specifies that the system must support multiple payment methods for consultation fee transactions.
3. **Non-Functional Requirement.** It defines a performance constraint by requiring appointment confirmations to be sent within 5 seconds.
4. **Non-Functional Requirement.** It specifies a scalability and performance requirement by supporting at least 20,000 concurrent users without noticeable degradation.
5. **Functional Requirement.** It describes a feature that sends real-time notifications to doctors when patients upload medical reports.
6. **Non-Functional Requirement.** It defines system reliability and availability by requiring at least 99.9% uptime and limiting maintenance downtime.

b) Suggest one additional functional requirement that could enhance the system and explain why it is useful.

- The system should allow patients to cancel or reschedule appointments before the scheduled appointment time.
- This improves convenience for both patients and doctors by keeping appointment schedules accurate and reducing missed appointments.

2.

Briefly explain the key practices of Extreme Programming (XP), such as Pair Programming, Test First Development, and Refactoring.

- **Pair Programming**
  - Two developers work together on the same code, frequently switching between the roles of driver (coder) and navigator (observer).
  - Continuous code review during development produces better quality software with fewer defects and lower maintenance cost.
  - Team members support each other and gain knowledge through collaboration.
- **Test First Development (TFD)**
  - Unit tests are written before the actual implementation code.
  - The development code is then written to make the tests pass.
  - This practice improves code quality, detects errors early, and ensures that each component meets its expected behavior.
- **Refactoring**
  - Refactoring is the process of improving the internal structure of existing code without changing its external behavior.
  - It includes removing duplicate code, splitting large classes or methods into smaller ones, and using meaningful variable names.
  - Refactoring improves software design, makes future requirement changes easier, and increases maintainability and extensibility.